

SetProbe: Description-First Relation-Wise Self-Consistency for Closed-Book Knowledge Base Construction[☆]

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Abstract

The AKBC 2026 shared task asks a language model to recover complete object sets closed-book. SetProbe anchors each subject, samples several candidate sets, and aggregates them per relation, reaching 0.513 macro F1. Sample count, aggregation rule, and abstention thresholds each move the score by less than the evaluation set’s own 0.045 confidence interval; only the choice of model does. The binding constraint is elementary: closed-book prediction cannot recover an object that appears in no sample.

1 Introduction

How can knowledge stored in a language model be turned into a complete database entry? A model may readily answer that the Dominican Republic borders Haiti, yet the same question becomes considerably harder when the expected answer contains nine neighboring countries, 80 award recipients, or no object at all. The AKBC 2026 shared task isolates this problem: given a subject s and a relation r , a system must predict the complete set of object strings $\{o_1, \dots, o_k\}$ (Kalo et al., 2026).

Consider a country with three gold neighbors. Returning two correct ones yields perfect precision and $2/3$ recall, while adding three plausible but wrong neighbors leaves recall unchanged and cuts precision to $2/5$; inventing a death city for a living person turns an otherwise perfect row into a complete miss. The design question is therefore not whether the model knows a fact, but when a further candidate should be admitted.

[☆]The views expressed in this paper are solely those of the author and do not necessarily reflect the views of the European Central Bank or the European System of Central Banks. All code and data for full replication are available at <https://github.com/JannicCutura/emnlp-kbc-2026>.

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We answer it with description-first relation-wise self-consistency, which identifies the exact subject, generates candidate sets through prompts exposing different recall paths, and admits empty answers into aggregation rather than gating generation beforehand. It reaches 0.513 macro F1 on validation.

The instructive result however is what did not move that number. Sample count is flat from three to twenty passes, no aggregation rule beats a plain median, and calibrating abstention thresholds on training traces reproduced the defaults exactly. Only the model mattered, for an elementary reason: a correct area value appears in the samples for 0.69 of rows under one eligible model and 0.43 under another, and no rule selects a value never generated.

2 Task

The dataset contains six relations and 477 training, 478 validation, and 475 test rows; we reserve validation for evaluation, and fit prompts and thresholds on training data only. Cardinality varies sharply across relations. Area and capacity carry exactly one numeric object, accepted within 5% relative tolerance. Death cities and stock exchanges are gold-empty on 39 and 36 of their 100 validation rows, so returning nothing is often the correct answer. Borders reach 14 neighbors and one award row has 610 recipients. String predictions are matched after normalization and alias expansion, and the official score is macro-averaged precision, recall, and F1 over subject–relation pairs.

The setting is closed-book: inference may use open-weight models totalling at most 32 billion published parameters, but may not query the web, a retrieval index, an external corpus, or a knowledge base, and fine-tuning is prohibited. In-context examples, multi-step inference, and non-neural post-processing are allowed (Kalo et al., 2026). Our system therefore uses the training split only for

few-shot demonstrations and no external factual source.

3 System

3.1 Common inference procedure

Figure 1 shows one row of inference. All calls use one open-weight model through an OpenAI-compatible chat endpoint. The model first produces a short identity anchor describing entity type, scope, and likely namesake confusions, then receives that anchor as explicitly uncertain context and generates n_r concise rationales and candidate sets. Each pass uses a different seed, recall focus, and rotating leave-one-subject-out sample of training demonstrations. Passes are independent and issued concurrently, with trace records following request order rather than completion order, so a concurrent run reproduces a serial one exactly.

For relations that permit empty answers, the demonstration mix mirrors the relation’s own training empty rate—two of five for death cities, one of five for borders—rather than being fixed in advance. An earlier version alternated empty and non-empty examples, showing a 60% empty rate for every relation and biasing the model toward abstention on precisely the relations where abstention is scored.

Candidate strings are parsed from the first valid JSON structure and normalized with the official evaluator’s rules. We additionally build relation-specific alias maps from the nested aliases in the training split, rewriting an alias only when all its training occurrences identify one canonical entity, before support is counted so that equivalent surfaces reinforce rather than split a vote. A response the parser cannot read is dropped from its row’s aggregate rather than counted as empty, which would convert a parser failure into a vote for “no objects”; this never occurred in 5,392 validation calls.

3.2 Relation-specific policies

Table 1 gives the pass count, aggregation rule, and post-processing for each relation; the prompts themselves are in the repository. Three of these choices are not self-evident.

Seven numeric samples, and a median. Both follow a training-only sweep rather than convention. Over the 100 training area rows accuracy is flat between 0.53 and 0.58 for every count from 3 to 20 and for every estimator we tried, including mode-seeking, trimmed, and order-of-magnitude-filtered variants. Twenty samples cost three times

Relation	n_r	Aggregation	After
Area	7	median	—
Capacity	7	median	—
Death city	20	plurality w/ empty	—
Exchange	10	support + empty	—
Borders	3	union	compl. + verif.
Awards	20	support union	compl.

Table 1: Per-relation policy. Post-processing runs only on a non-empty aggregate. Support thresholds are 0.30 for exchanges and 0.05 for awards, and the empty-vote threshold is 0.50 for both abstention relations.

as much as seven and bought nothing measurable; the median survives by default rather than by merit.

Post-processing runs only on a non-empty set.

This gate was the largest correction we made and we did not anticipate it. Asking “which neighbors are missing?” is a leading question, and a model asked for omissions supplies them whether or not any exist: in 12 of the 18 gold-empty validation rows every sample correctly returned no neighbor, after which the completion pass invented Italy for Malta, Cuba and the United States for the Bahamas, and three maritime neighbors for Australia. The verifier accepted all of them. Across the split completion introduced 36 neighbors while verification removed 22. An omission pass presupposes a set to have omissions from; gating it raised border F1 from 0.796 to 0.859.

Awards are partitioned by era. A single answer may contain 610 recipients, so one undifferentiated prompt invites both truncation and omission. Twenty passes cycle through five time windows and a sixth focus organized by profession and recipient type. We apply no binary verifier: an earlier one cut correct recipients from 96 to 56.

4 Experimental Setup

Models. We evaluate Gemma 4 31B-IT (30.7B published total parameters) and Gemma 3 27B-IT (27.4B) on complete splits, and screen Qwen3.5 27B on area alone. Every relation in a run uses one model, so the inference-time total never exceeds 32B; quantization does not reduce it and mixture-of-experts models count by published total.

Inference configuration. Inference runs through a hosted OpenAI-compatible gateway. Such gateways route one model name across backends at precisions from fp4 to bf16, so we pin the provider and bf16 precision and record both on every call;

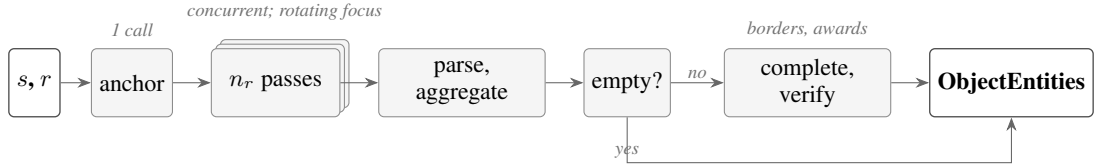


Figure 1: One row of inference. Completion and verification run only on a non-empty aggregate: asking an omission pass what is missing from an empty set invents members.

Relation	G4 31B	G3 27B	Δ
Land borders	.989	.859	+.130*
Stock exchange	.713	.648	+.064
Death city	.460	.340	+.120
Capacity	.220	.230	-.010
Awards	.148	.205	-.057
Area	.370	.520	-.150*
All relations	.513	.490	+.022

Table 2: Per-relation validation F1 for the two models. * marks differences whose 95% paired-bootstrap interval excludes zero. The overall difference does not ($p = 0.33$).

otherwise a comparison confounds model identity with numeric precision. Runs use temperature 0.6, top- p 0.95, a 4,096-token cap, and five in-context examples except for awards. Hidden reasoning is disabled explicitly: it is not bounded by the generation cap and is billed at the completion rate, and one 100-row screen on a reasoning-tuned model emitted 3.87M such tokens. A complete validation run costs 5,392 calls and about 80 minutes.

Selection and ablations. All thresholds are selected from complete leave-one-out training traces with the official evaluator, and the locked configuration is then evaluated on the validation rows. We do not tune thresholds, sample counts, aggregation rules, or entity-specific rules against validation answers, and because the validation gold is distributed with the task we additionally report training-split scores as an independent estimate. The historical baseline is an earlier GPT-OSS-20B prototype scoring 0.460, whose hard empty-answer gates and award verifier the present system removes. Comparisons use a paired bootstrap over rows with 20,000 resamples.

5 Results

On all 478 validation rows the system reaches macro precision 0.542, recall 0.578, and F1 0.513, against 0.460 for an earlier gated pipeline of ours on a different model. The 95% bootstrap interval

is $[0.468, 0.557]$, and an independent run on the training split scores 0.484.

Table 2 shows the two models are complementary rather than ordered. Gemma 4 is better on every relation rewarding adherence to a precise definition, reaching 0.989 on borders where one row separates it from a perfect score, while Gemma 3 is better on the two relations rewarding raw numeric recall. Only those two effects survive a paired bootstrap and they nearly cancel: the overall $+0.022$ has interval $[-0.022, +0.067]$ and $p = 0.33$. The best system we can describe is one we may not submit, since pairing Gemma 3’s area with Gemma 4’s other relations would score 0.544 but sum to 58.1B parameters.

Tuning does not move the score; knowledge does.

Three of our four controlled comparisons came back empty. Sample count is flat from three to twenty passes; no aggregation rule we tried improves on a plain median; and sweeping the abstention thresholds over complete training traces simply rediscovered the values we had guessed, leaving death and exchange where they already stood. Only the model moved anything, and it moved relations in opposite directions rather than lifting the whole.

The reason is visible in the candidate pools. For `hasArea` a correct value appears in at least one sample for 0.69 of rows under Gemma 3 and 0.43 under Gemma 4, while the aggregates return 0.54 and 0.37; for `hasCapacity` the ceiling is 0.19. Where the correct value is absent from every sample no selection rule can recover it, and where it is present but in the minority, agreement-based aggregation discards it because an incorrect value dominates the distribution.

6 Error Analysis

Abstention is too conservative, not too aggressive.

On stock exchanges the system predicts empty on all 36 gold-empty rows with no spurious answer at all, yet returns nothing on 26 rows that do have a listing; death cities are correct on

33 of 39 gold-empty rows against 32 rows where a recoverable city was suppressed. This is a negative result about our own design: the prototype’s hard pre-generation gate suppressed 28 of 61 non-empty death rows, and the generation-first voting that replaced it suppresses 32. The mechanism changed from a classifier to a vote but the operating point did not, because the samples agree on silence—when the model does not know, twenty passes independently return nothing, and no rule over unanimous abstentions manufactures a city.

Parsing is not the limiting factor. Because our central claim is that correct objects are absent rather than mishandled, we audited it: across 5,392 calls the parser never dropped a gold alias present in an answer array. Of the incorrect rows, 58 of 63 for area, 71 of 78 for capacity and 38 of 48 for death contain no correct value anywhere in any response, while aggregation discards a correct minority value in 15 further rows, worth about 0.03 overall.

Stock exchanges are the exception. In 9 of 27 incorrect rows the model names the correct exchange in its rationale and then returns an empty set because it believes the listing has ended—Leclanché is “previously listed on the SIX Swiss Exchange, but delisted following its acquisition”. Given that judgement it correctly obeys our instruction to return nothing for a company that is not independently public, so reading the answer out of the rationale would mean admitting its own negations. Here the ceiling is set by temporal belief rather than by missing knowledge.

Remaining errors. Median numeric error is 3.2%, so failures are misidentification rather than noise: *Mainland* returns 9,596,961, the area of China rather than of the 540 km² island. The identity anchor was introduced for exactly this and does not eliminate it. Border recall is 0.998 once the omission pass is gated. Awards average 68.0 predicted entities against a gold mean of 145.8, up from 15.2, but ten rows cannot be measured to better than ± 0.11 .

7 Related Work

LAMA established that pretrained models hold retrievable factual associations (Petroni et al., 2019) and closed-book question answering that they can be queried without document access (Roberts et al., 2020). Both ask whether a known object is recovered, not how many exist, which is what makes this

task hard: a probe scored on its top answer never has to decide that a country has nine neighbors or that a living person has no city of death. Phrasing alone moves probing results substantially (Elazar et al., 2021), so we vary phrasing across passes rather than search for one best prompt. Retrieval would supply exactly the long-tail facts parametric memory covers worst (Lewis et al., 2020; Mallen et al., 2023); forbidding it leaves prompt diversity and repeated sampling as the only sources of variation.

The task has run annually with shifting resource rules (Kalo et al., 2024, 2025). The 2025 edition is closest to ours: it fixed a single small model, so methods rather than models distinguished the entries. The winner combined chain-of-thought prompting with relation-wise self-consistency and train-tuned per-relation aggregation (Albert-Roulhac and Zouaq, 2025); the runner-up generated an entity description before extraction and decomposed high-cardinality relations (He and Razniewski, 2025). We adopt both. Our contribution is not their invention but a measurement of what they are worth once the model may vary, which a fixed-model setting cannot reveal.

Self-consistency aggregates repeated samples by majority vote (Wang et al., 2023), but assumes one final answer; set prediction instead votes over normalized candidates, where a value generated once in twenty passes may still be correct. Chain-of-verification reduces hallucination through separate checking questions (Dhuliawala et al., 2024), though a closed-book verifier consults the same parameters as the generator, so we verify only borders. Finally, a third of death and exchange rows are gold-empty, making abstention a first-class prediction as in unanswerable-question benchmarks (Rajpurkar et al., 2018) and calibrated selective prediction (Kamath et al., 2020).

8 Conclusion

We presented a description-first self-consistency pipeline for closed-book knowledge base construction, sharing one orchestration pattern across relations while assigning aggregation by cardinality. It scores 0.513 on validation.

The more useful result is where the remaining error lives. Varying sample count, aggregation rule, abstention threshold, and model, only the last changes the score, and its gains are relation-specific and mutually exclusive under a parameter budget.

An incorrect value that dominates the sample distribution defeats agreement-based selection precisely when agreement is what one would like to trust. Effort is therefore better spent on elicitation diversity than on aggregation, and reporting a candidate-pool ceiling alongside the score would make systems easier to compare.

Limitations

The system cannot recover facts absent from the model’s pretraining data, which matters most for deaths, listings, and awards close to the July 2026 reference date. Reusing one model for border generation and verification may reinforce rather than correct a misconception, and train-calibrated thresholds may shift under distribution change. We therefore reserve the test split for a single final evaluation and record every prompt, response, and configuration.

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